

# Corrosion Resistant Biopolymer Coatings

Developing a method to characterize the properties of polycannabinoids in transient electronics applications

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## Overview

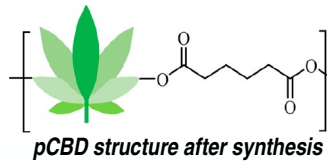
E-waste is the fastest growing solid waste stream in the world (WHO)

Transient electronics are designed to operate over a typically short and predefined duration of time

Polycannabinoid properties:

- Hydrophobic
- Naturally antioxidative
- Degradable (no additives)

Properties tested: Water Absorption Test, Resistance Test



## Methods

Original methodology for resistance test

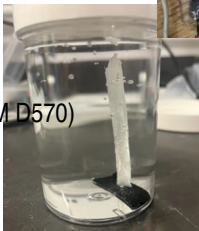
- ☐ Flower of Sulfur Test ASTM B809-95
- ☐ Exposing conductive traces to sulfur vapor at high temperature accelerates the oxidation process from several years to several days, allowing high-throughput testing of different barrier coatings



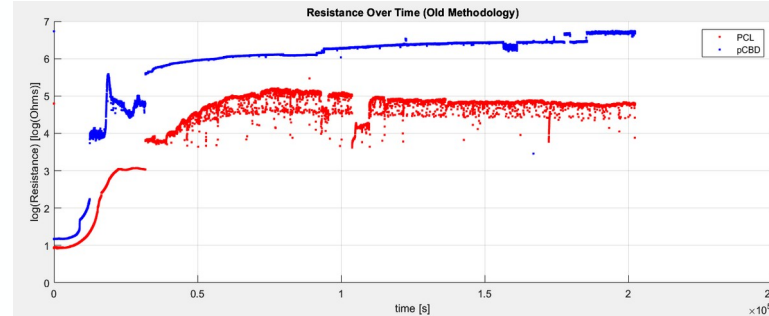
Changes made to method:

- ☐ Increasing test efficiency
- ☐ Standardizing- thickness of coatings
- ☐ New testing setup

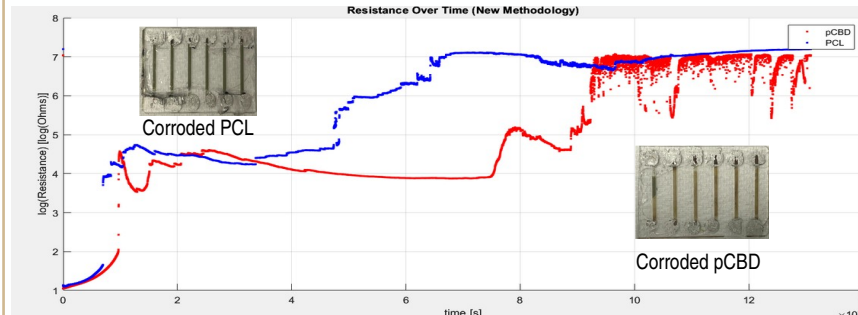
Water absorption test (referenced ASTM D570)



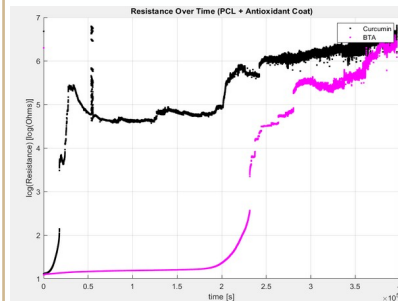
## Results



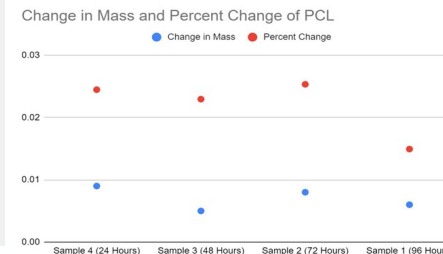
PCL vs. PCBD Resistance test with the old method



PCL vs. PCBD Resistance test with the improved method



PCL and Antioxidants Resistance test with the improved method



PCL Water Absorption Test

## Discussion

- Expected ~2% change for water absorption test, limiting factor is accuracy of weighing scale
- Expected smooth, linear curve for resistance tests
- Silver epoxy method allows for greater control and easier troubleshooting

## Conclusions

- Data indicates pCBD coat delayed corrosion by approximately 75% more time compared to just PCL
- On average, 2-4 samples out of every 6 showed disconnection
- Water absorption test requires a larger volume of samples, and heavier samples for consistent results
- Additional tests should be conducted to determine viability

## Future Work

- Run additional standardized tests
  - Tensile Test
  - Water Vapor Rate Test
  - Resistance Test in Ambient Environment
- Improve stability of resistance test equipment
- Compare the effects of antioxidants to pCBDs' antioxidant properties
- Test pCBD coating on Copper inks
- Apply pCBDs to printable electronics and transient electronics
- Utilize FTIR to identify patterns in corrosion progress